

Clinical Knowledge Assistant: System Architecture & Requirements

1. Clinical Context & Purpose

A RAG-powered clinical decision support assistant designed to retrieve verified medical literature, clinical practice guidelines (ADA, AHA, IDSA, CDC), and hospital formularies to assist clinicians in fast, evidence-based decision-making.

2. Critical Regulatory & Safety Requirements

Requirement Area	Specification Standard	Technical Implementation
Strict Evidence Grounding	Answers derived exclusively from indexed medical literature	High vector similarity threshold + citation enforcement
Mandatory Clinical Disclaimer	Every output includes standard non-prescriptive medical notice	Automated post-processing disclaimer injection
HIPAA PHI Protection	Zero Protected Health Information (PHI) transmitted to external APIs	Client-side Presidio / spaCy de-identification
Source Traceability	Every claim must map to a specific guideline document and paragraph	Structured citation metadata output
Query Latency	Guideline retrieval and synthesis completed in under 3 seconds	ChromaDB vector indexing + GPT-3.5-Turbo